PRFAQ-Proposal-SILO

Identify GSF member and non-member organizations relevant to data centre infrastructure and sustainability interoperability

- Cloud providers / hyperscalers: Google, Microsoft, IBM
- Cooling, power and facility infrastructure vendors: Schneider Electric, Siemens, ZutaCore (members); Vertiv, Eaton, nVent, Munters, Stulz (non-members)
- Infrastructure and ontology bodies: Sustainable Digital Infrastructure Alliance, Boavizta (members); DMTF (Redfish), Open Compute Project, ASHRAE TC 9.9, Project Haystack, Brick Schema (non-members)
- Carbon and grid data providers: Electricity Maps, WattTime, Energy Web
- Colocation and data centre operators: NTT DATA (member); Equinix, Digital Realty, CyrusOne (non-members)
- Enterprise operators of large estates: UBS, Goldman Sachs, Bloomberg, Mastercard, Amadeus
- Research and public bodies: NREL, Chalmers University of Technology, University of Edinburgh, GovTech Singapore, IMDA

Define the key personas we need to serve, including data centre operators and facility infrastructure vendors

- 1. The Data Centre Operator (The Consumer)
  - This persona represents facility engineers, critical infrastructure managers and operations leads running mixed-vendor estates.
  - Their primary goal is to run adaptive cooling, dynamic workload placement and predictive maintenance across the whole estate without commissioning a bespoke integration for every vendor at every site.
- 2. The Facility Infrastructure Vendor (The Enabler)
  - This persona includes cooling, power, rack and sensing manufacturers, and their firmware and telemetry teams.
  - Their goal is to expose their telemetry once, in a form that any platform can consume, without disclosing proprietary internals or maintaining a separate integration per customer.
- 3. The Platform and Optimisation Engine Builder (The Integrator)
  - This persona represents DCIM, BMS, orchestration and optimisation software vendors.

  - Their goal is to build once against a common interface and deploy across heterogeneous infrastructure, rather than absorbing per-vendor normalisation as a permanent engineering cost.

- 4. The Sustainability Lead (The Stakeholder)
  - While not a direct technical user of the telemetry layer, this persona depends entirely on the data it produces.
  - Primary Goal: To obtain consistent, comparable and auditable PUE, WUE, CUE, energy, water and carbon data for EED and CSRD reporting.
- 5. The Grid / Energy Optimisation partner (The Coordinator)
  - Represents utilities, aggregators and energy optimisation platforms.
  - Their goal is to align grid needs with data centre flexibility and performance constraints.

Draft a short PR/FAQ document for a SILO specification guideline

PRFAQ: Sustainability Interoperability Layer for Operations (SILO)

Press Release

Standardizing Cross-Domain Sustainability Telemetry with SILO

Project Proposal: Sustainability Interoperability Layer for Operations

Problem: Operational telemetry from compute, power, cooling, networking, storage and facility systems arrives in incompatible vendor schemas with no shared meaning, so cross-domain sustainability optimisation requires a bespoke integration per vendor per site. This also limits the ability to align data centre performance with grid needs, translating external energy and carbon signals into coordinated compute, power and cooling actions while respecting workload, thermal and reliability constraints.

Solution: Define an open, vendor-neutral semantic layer above the existing WDPC Data Bus, providing canonical semantic models, standardised APIs, normalisation rules and governance so any platform can consume telemetry from any vendor through one interface, and relate operational capabilities and performance constraints to external grid and energy signals

Key Deliverables:

- Canonical semantic models: cross-domain entities, relationships, taxonomies and engineering units, including operational capabilities, performance constraints and their relationship to grid and energy signals.

- Standardised APIs for discovery, query and streaming of real-time and historical operational data.
- Interoperability framework: normalisation rules and schema validation for vendor-specific telemetry.
- Governance model: SILO Technical Steering Committee, metric approval process, versioning aligned to WDPC releases.

Next Steps for Steering Committee:

- Approve launch of a [4-6] week Assembly to agree scope, ontology structure and governance.
- Confirm Assembly type: Private (members only) or Public (non-members by invitation) — recommendation is Public, to reach the facility and cooling vendor community not yet represented in membership.
- Formal vote on project commencement scheduled for [DATE].

LONDON, UK — The Green Software Foundation (GSF) announces the launch of the **"SILO"** project, an initiative to give data centre operators a single, vendor-neutral language for operational sustainability data by defining a standardized semantic interoperability layer above the Foundation's Workload-Dynamic Power and Cooling (WDPC) specification.

As regulatory pressure moves from voluntary disclosure to mandatory performance, the absence of comparable cross-domain telemetry has become the binding constraint. Under the EU Energy Efficiency Directive (2023/1791), every data centre above 500 kW of IT power demand already files annual sustainability returns, and Article 12(5) obliges the Commission to assess that data and introduce minimum performance standards where appropriate. Operators cannot manage what they cannot compare, and today a rack inlet temperature, a pump flow rate and a node power draw arrive from three vendors in three schemas, three units and three notions of which asset they describe.

"The industry has spent a decade optimising PUE inside individual systems. The remaining efficiency is in the coordination between them, and coordination requires shared meaning. SILO is the vocabulary that lets a cooling system, a workload scheduler and a reporting tool describe the same facility the same way." — GSF Spokesperson

The project builds directly on WDPC and on Project Mycelium, the 2025 Assembly convened to break hardware and software silos for end-to-end sustainability, which established that cross-domain signalling is tractable but requires an agreed semantic foundation to scale beyond single-vendor environments. It is complementary to the Foundation's white paper on the external grid interface, *Bridging the Sustainability Gap*: that specification governs how a facility talks to the grid, while SILO governs whether the facility's own systems can produce a coherent answer to send.

Frequently Asked Questions

1. What is the SILO project?

An open specification defining a vendor-neutral semantic layer for operational sustainability telemetry inside the data centre. It comprises four components: canonical semantic models giving common meaning and units to metrics across IT and facility domains; standardised APIs for discovering, querying and streaming that data; an interoperability framework normalising vendor-specific telemetry and validating it for schema compliance and quality; and a governance model maintained by a SILO Technical Steering Committee with versioning aligned to WDPC releases.

2. Why a semantic layer instead of another telemetry protocol?

The protocols already exist and work. What does not exist is agreement on what their outputs mean in relation to one another. SILO is protocol-agnostic and plugs into the existing WDPC Data Bus; it adds interpretation, not transport. Nothing in the current stack is replaced.

3. What are the primary deliverables?

Deliverable	Description
Semantic Models	Vendor-neutral ontology defining meaning, relationships, taxonomies and engineering units across compute, power, cooling, networking, storage and facility domains
APIs	Standardised interfaces for discovery, query and streaming of real-time and historical operational data
Interoperability Framework	Normalisation rules translating vendor telemetry into the canonical format, with schema and quality validation
Governance	SILO Technical Steering Committee, formal metric approval process, versioning aligned to WDPC

4. How will this help operators reduce energy, water and carbon?

By making cross-domain optimisation buildable rather than bespoke. Standardised access to workload, power, thermal, fluid, environmental and facility metrics supports adaptive cooling, dynamic workload placement, energy efficiency optimisation, predictive maintenance, heat reuse optimisation and automated sustainability reporting. Success is measured on the availability, consistency and interoperability of PUE, WUE, CUE, energy, water and carbon metrics across participating systems — that is, whether an operator can compare like with like across a fleet and demonstrate improvement over time. It also

connects grid signals with data centre flexibility and performance constraints, enabling safe adaptation of workloads and infrastructure while measuring sustainability and performance impact.

5. Comparison of Existing Standards and the Remaining Gap

Standard	Covers	Domain	Reference
DMTF Redfish / Swordfish	Server and storage hardware management	IT hardware	DMTF Redfish
BACnet / Modbus	Building and facility system control	Facility systems	Existing industry protocols
Project Haystack / Brick Schema	Facility asset and equipment modelling	Buildings	Project Haystack
GSF WDPC	Workload-to-facility power and cooling signalling	Cross-domain, single-vendor viable	GSF WDPC
OpenTelemetry semconv	Software observability telemetry	Software	OpenTelemetry
Gap addressed by SILO	Cross-domain semantic normalisation and mapping across all of the above	Multi-vendor estates	GSF SILO
GSF External Grid Interface	Data centre-to-grid signaling and coordination	Grid / Energy	GSF - Bridging the Sustainability Gap

SILO maps across these standards rather than competing with them. Reuse over reinvention is an explicit design constraint: the Assembly's first task is deciding which existing models SILO adopts.

6. What is the biggest risk to the project?

Ontology scope creep. SILO nominally spans six domains, any one of which could absorb an Assembly on its own. The mitigation is to constrain the first blueprint to the minimum viable ontology supporting three named use cases — adaptive cooling, dynamic workload placement and automated PUE/WUE/CUE reporting — and route all further metric types

through the governance process afterwards. This should be written into the scope statement rather than left to participants' restraint.

Presentation Slide: SILO Project Overview

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Key Deliverables:

- Canonical semantic models for cross-domain entities, relationships and engineering units.
- Standardised APIs for discovery, query and streaming of operational data.
- Interoperability framework with normalisation rules and schema validation.
- Governance model: SILO Technical Steering Committee, versioning aligned to WDPC.

Next Steps for Steering Committee Approval:

- Approve launch of a [4-6] week Assembly to agree scope, ontology structure and governance.
- Confirm Assembly type: Private or Public, with Public recommended for facility vendor reach.
- Formal vote on project commencement scheduled for [DATE].